Semantic Clustering of Agricultural Drone Imagery Using Vision-Language Embeddings

Research Notes - Draft v0.2.2

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Abstract

This study explores the application of vision-language embedding models for automatic semantic categorization of agricultural drone imagery. Using CLIP (Contrastive Language-Image Pre-training) embeddings with UMAP dimensionality reduction, we demonstrate that drone survey images can be automatically clustered by semantic content—specifically by flight altitude and imaging purpose—without manual labeling. Analysis of 97 drone images from a tomato field survey in Santa Ines, Chile reveals two distinct clusters: close-up crop inspection images (92%) and high-altitude field mapping images (8%). Similarity network analysis confirms strong cluster cohesion with only 2.7% cross-category connections. These findings suggest that pretrained vision-language models offer a promising approach for organizing and filtering large agricultural drone datasets, potentially reducing manual curation effort in precision agriculture workflows.

[TODO: Refine abstract after completing results section**]**

Contents

1 Introduction

1.1 Background

Unmanned Aerial Vehicles (UAVs) have become essential tools in precision agriculture, enabling high-resolution monitoring of crop health, irrigation patterns, and field conditions. Modern drone surveys generate large volumes of imagery—often hundreds to thousands of images per flight session—creating significant data management challenges.

[TODO: Add citations: drone imagery in agriculture, data volume challenges**]**

A typical agricultural drone survey may include:

- **Systematic mapping flights**: High-altitude images for orthomosaic generation
- **Targeted inspection flights**: Low-altitude close-ups of specific crop areas
- **Oblique/transitional images**: Various angles during ascent/descent

Organizing this heterogeneous imagery typically requires manual review or relies solely on GPS metadata, which captures *where* images were taken but not *what* they contain semantically.

1.2 Vision-Language Models for Image Understanding

Recent advances in vision-language models, particularly CLIP (Contrastive Language-Image Pre-training), have demonstrated remarkable zero-shot capabilities for image understanding. These models learn joint embeddings of images and text, enabling semantic similarity comparisons without task-specific training.

[OBS: CLIP embeddings capture semantic content (altitude/purpose) rather than just spatial position—this is a key distinction from GPS-based organization**]**

1.3 Research Gap

While vision-language models have been extensively studied for general image classification and retrieval, their application to agricultural drone imagery organization remains underexplored. Specifically:

1. Can pretrained models (without agricultural fine-tuning) meaningfully cluster drone imagery?
2. What semantic distinctions do these models capture in agricultural contexts?
3. How do embedding-based clusters compare to traditional metadata-based organization?

[TODO: Expand literature review: existing work on drone image organization, CLIP applications in remote sensing**]**

2 Objectives

2.1 General Objective

To evaluate the effectiveness of vision-language embedding models for automatic semantic organization of agricultural drone imagery, with application to precision agriculture data management workflows.

2.2 Specific Objectives

1. **Embedding Analysis**: Compute and visualize CLIP embeddings for drone survey imagery to identify natural semantic clusters.
2. **Cluster Characterization**: Analyze the semantic distinctions captured by embedding-based clusters (e.g., altitude, crop visibility, image purpose).
3. **Comparison with Metadata**: Compare embedding-based organization with GPS/EXIF metadata-based approaches to understand complementary information.
4. **Similarity Search**: Implement and evaluate content-based image retrieval for finding similar drone images within large datasets.
5. **Workflow Integration**: Develop practical tools for integrating embedding-based organization into agricultural data pipelines.

2.3 Hypotheses

H1 Pretrained CLIP embeddings will separate drone images by semantic content (flight altitude, imaging purpose) without domain-specific training.

H2 Embedding-based clusters will capture information orthogonal to GPS position—images from different locations but similar purposes will cluster together.

H3 Similarity search based on embeddings will retrieve semantically relevant images more effectively than spatial proximity alone.

[TODO: Refine hypotheses based on preliminary results**]**

3 Materials and Methods

3.1 Study Area and Data Collection

3.1.1 Location

Santa Ines agricultural field, Chile. Tomato cultivation site monitored via drone surveys. **[TODO:** Add exact coordinates, field size, crop details**]**

3.1.2 Drone Imagery Dataset

- **Platform**: DJI drone (model TBD)
- **Image count**: 97 images
- **File format**: JPEG, approximately 10-12 MB per image
- **Resolution**: High-resolution (dimensions TBD from metadata)
- **Date**: January 2026

3.2 Software Environment

3.2.1 Dataset Management

FiftyOne v1.12.0 (?) was used for dataset management, embedding computation, and visualization. FiftyOne provides:

- Efficient handling of large image datasets
- Built-in integration with embedding models
- Interactive visualization tools
- Similarity search functionality

3.2.2 Analysis Tools

- **Python 3.12**: Data processing and embedding computation
- **R 4.3.3**: Statistical analysis and visualization (ggplot2)
- **PyTorch**: Deep learning backend for CLIP model

3.3 Embedding Computation

3.3.1 Model Selection

CLIP ViT-B/32 (Vision Transformer, Base, 32x32 patches) was selected for embedding computation. This model:

- Provides 512-dimensional embeddings per image
- Was pretrained on 400M image-text pairs
- Offers good balance of computational efficiency and semantic richness

3.3.2 Dimensionality Reduction

UMAP (Uniform Manifold Approximation and Projection) was applied to reduce embeddings from 512D to 2D for visualization:

$$\mathbf{z}_i = \text{UMAP}(\mathbf{e}_i), \quad \mathbf{e}_i \in \mathbb{R}^{512}, \quad \mathbf{z}_i \in \mathbb{R}^2 \tag{1}$$

[TODO: Add UMAP hyperparameters: n_neighbors, min_dist, metric**]**

3.3.3 Implementation

Listing 1: Embedding computation with FiftyOne

```
import fiftyone as fo
import fiftyone.brain as fob

dataset = fo.load_dataset("santa_ines_dron")

# Compute CLIP embeddings with UMAP projection
fob.compute_visualization(
    dataset,
    brain_key="clip_viz",
    model="clip-vit-base32-torch",
    method="umap"
```

```
)

# Compute similarity index for retrieval
fob.compute_similarity(
    dataset,
    brain_key="clip_similarity",
    embeddings="clip_viz"
)
```

3.4 Clustering and Classification

Image clusters were identified through visual inspection of the UMAP projection. Two distinct clusters emerged:

1. **Main cluster** (89 images): Concentrated in UMAP region $x \in [6.8, 10.5]$, $y \in [0.5, 4.5]$
2. **Outlier cluster** (8 images): Isolated at $x \approx 11.5$, $y \approx -0.7$

Images were tagged based on cluster membership for subsequent analysis.

3.5 Similarity Network Analysis

To quantify cluster structure, a k-nearest neighbor graph was constructed:

- Each image connected to its $k = 3$ nearest neighbors in embedding space
- Distance metric: Euclidean distance in 2D UMAP space
- Edge classification: within-category vs. cross-category connections

3.6 Statistical Analysis

Cluster cohesion was evaluated using:

- Mean distance to k nearest neighbors
- Inter-cluster centroid distance
- Proportion of cross-category edges in similarity network

All statistical analyses and visualizations were performed in R using ggplot2.

4 Results

4.1 Combined Spatial and Visual Analysis

Figure 1 presents an integrated view of the Santa Ines drone survey analysis, demonstrating the fundamental difference between physical GPS locations and visual semantic clustering.

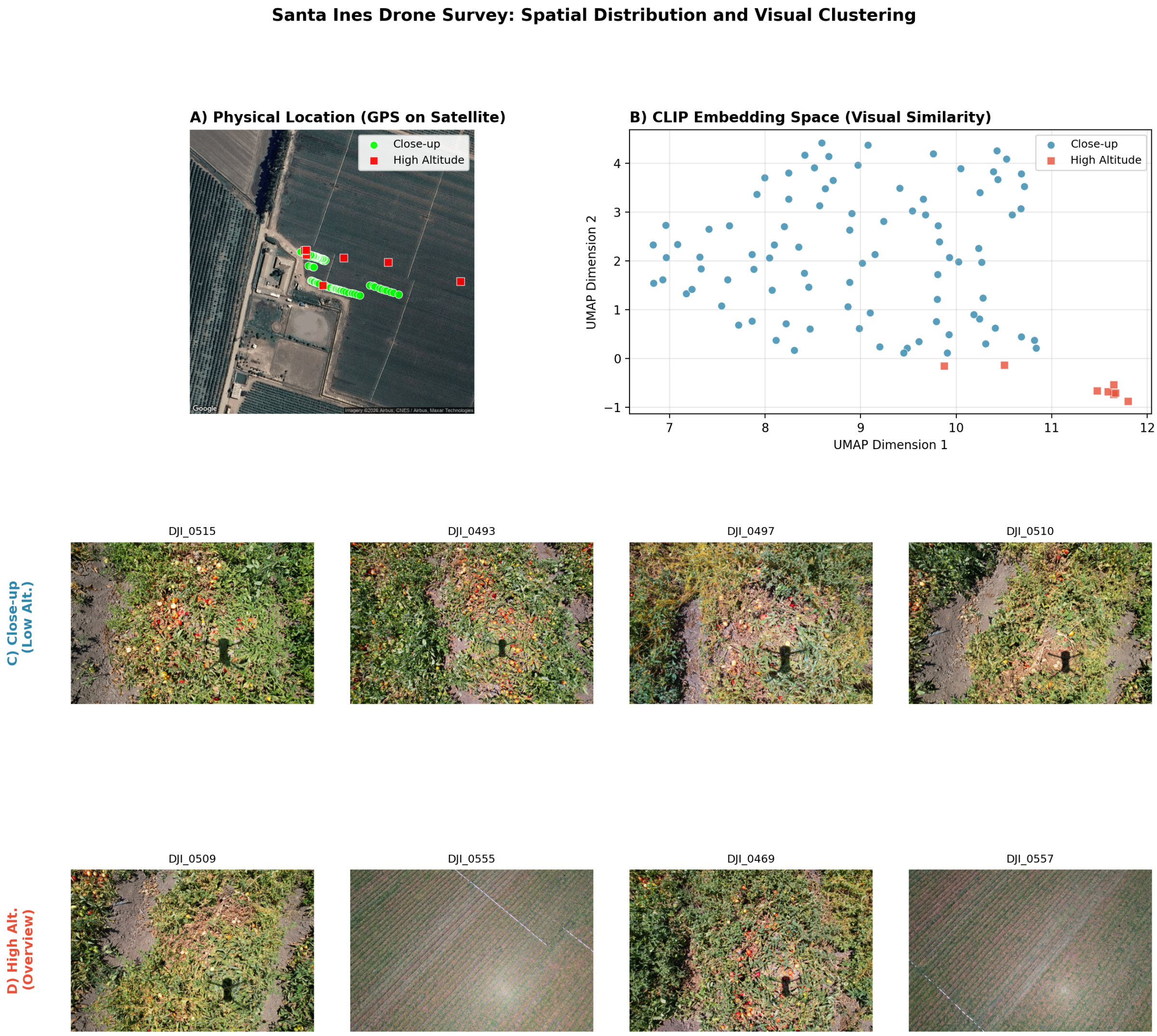

Figure 1: Combined analysis of Santa Ines drone imagery. **(A)** GPS coordinates overlaid on satellite imagery showing flight coverage; **(B)** UMAP projection of CLIP embeddings revealing two semantic clusters; **(C)** Representative close-up images capturing detailed crop inspection views; **(D)** Representative high-altitude overview images showing vineyard row patterns.

4.1.1 GPS vs Visual Embedding: Key Differences

Panel A reveals the physical flight pattern over the Santa Ines vineyard. The drone followed systematic transects across the crop rows, with close-up images (green markers) distributed along parallel flight lines and high-altitude overview images (red markers) captured at strategic positions across the field. Geographically, both image types are **spatially intermixed**—high-altitude shots were taken from various positions throughout the survey area.

In contrast, Panel B shows the CLIP embedding space where images cluster by **visual content rather than geographic location**. Despite being captured from scattered GPS positions, all high-altitude images form a tight cluster in the lower-right corner of the embedding space, clearly separated from the close-up images. This demonstrates that vision-language models organize images by semantic similarity (what the image depicts) rather than spatial proximity (where it was captured).

4.1.2 Visual Characteristics of Each Cluster

Panels C and D provide visual evidence for why CLIP separates these image categories:

Close-up images (Panel C) are characterized by:

- High detail of individual vine plants and foliage
- Visible leaf texture, color variations, and plant structure
- Limited field-of-view showing 1-3 plant rows
- Rich color palette dominated by greens and earth tones

High-altitude images (Panel D) exhibit distinct visual patterns:

- Geometric vineyard row patterns visible from above
- Uniform texture showing repetitive agricultural structure
- Wide field-of-view capturing dozens of rows
- Reduced color variation with dominant green/brown striping

[FIND: CLIP embeddings successfully capture the semantic distinction between "detailed crop inspection" and "field overview" image types, enabling automatic categorization based purely on visual content**]**

[OBS: This spatial-semantic decoupling has practical implications: images can be organized by content type regardless of capture location, enabling more intuitive dataset browsing and quality control workflows**]**

4.2 Embedding Space Visualization

CLIP embeddings projected to 2D via UMAP revealed two distinct clusters in the Santa Ines drone dataset (Figure 2).

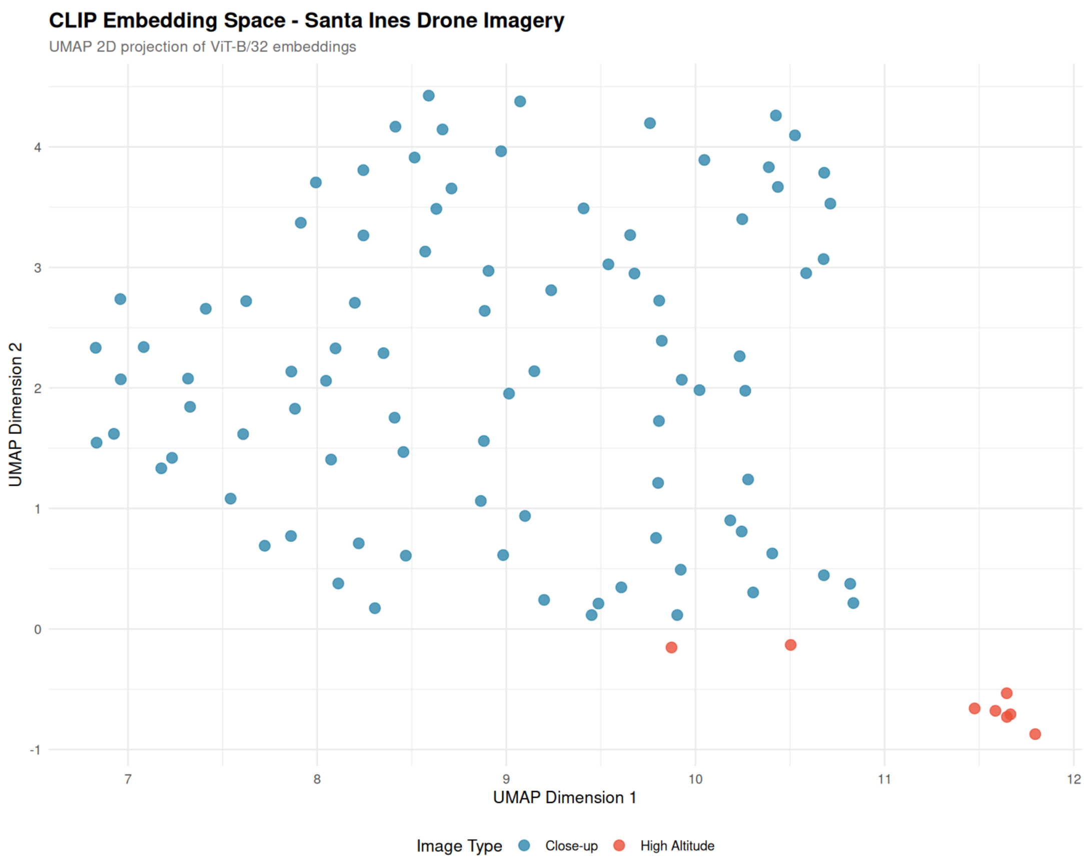

Figure 2: UMAP projection of CLIP embeddings for 97 drone images. Blue points represent close-up crop inspection images; red points represent high-altitude overview images.

4.2.1 Cluster Characteristics

Table 1: Summary statistics for identified image clusters

Category	Count	Percentage	UMAP X (mean ± sd)	UMAP Y (mean ± sd)
Close-up (crop inspection)	89	91.8%	8.97 ± 1.15	2.15 ± 1.28
High-altitude (overview)	8	8.2%	11.30 ± 0.70	-0.56 ± 0.27

[OBS: The high-altitude cluster shows lower variance (tighter grouping), suggesting more homogeneous visual content among overview images**]**

4.2.2 Inter-cluster Distance

The Euclidean distance between cluster centroids in UMAP space was **3.56 units**, indicating clear separation between semantic categories.

4.3 Similarity Network Structure

Network analysis of the 3-nearest-neighbor graph revealed strong within-cluster connectivity (Figure 3).

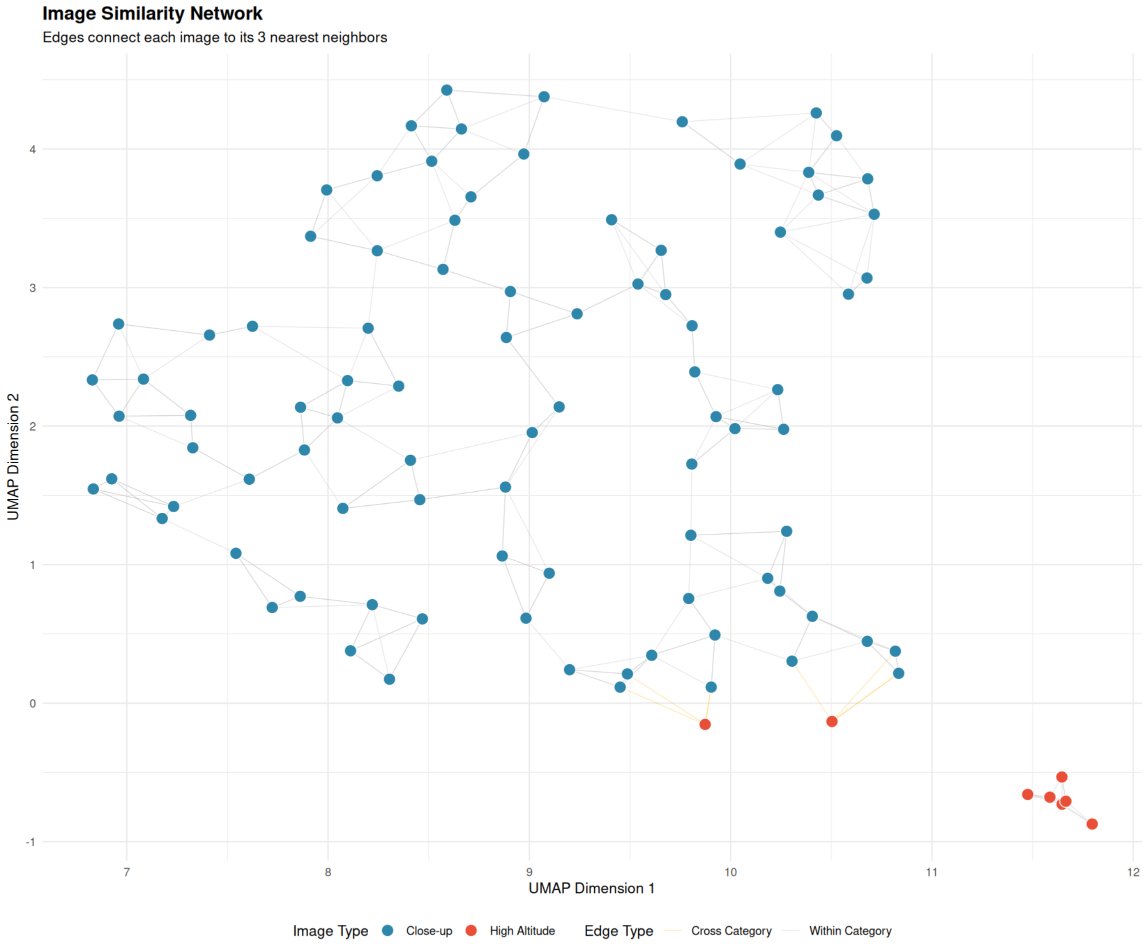

Figure 3: Image similarity network. Nodes represent images (blue = close-up, red = high-altitude). Gray edges connect images within the same category; yellow edges indicate cross-category connections.

4.3.1 Network Statistics

- **Total edges**: 291 (3 per image × 97 images)
- **Within-category edges**: 283 (97.3%)
- **Cross-category edges**: 8 (2.7%)

[FIND: Only 2.7% of nearest-neighbor connections cross category boundaries, confirming robust cluster separation**]**

4.4 Cluster Cohesion Analysis

Mean distance to 5 nearest neighbors was computed for each category (Figure 4).

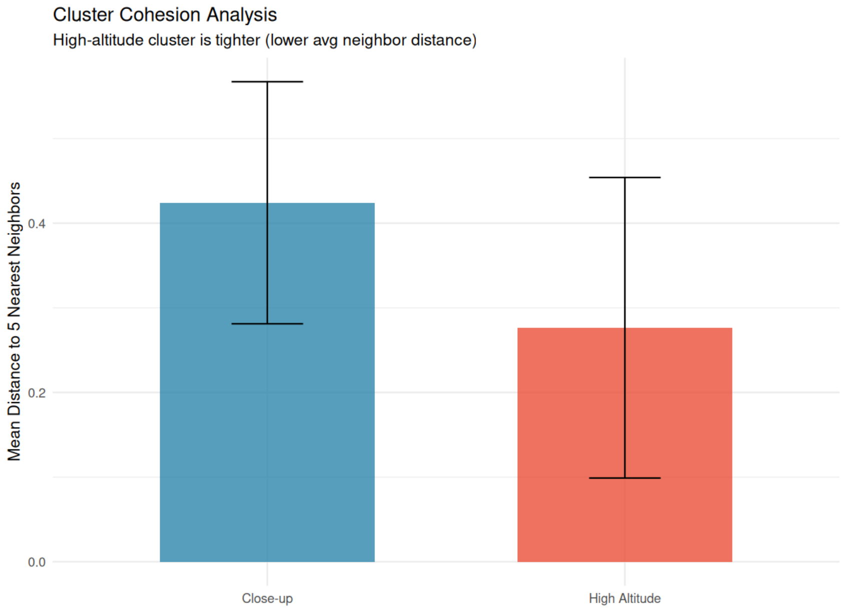

Figure 4: Cluster cohesion measured by mean distance to 5 nearest neighbors. Error bars show standard deviation.

Table 2: Cluster cohesion statistics

Category	**Mean Distance**	**Std Dev**	**n (neighbor pairs)**
Close-up	0.424	0.143	445
High-altitude	0.276	0.178	40

[OBS: High-altitude cluster is 35% more cohesive (lower neighbor distance) than the close-up cluster, despite having fewer samples**]**

4.5 Image Size Analysis

File size distribution showed modest differences between categories (Figure 5).

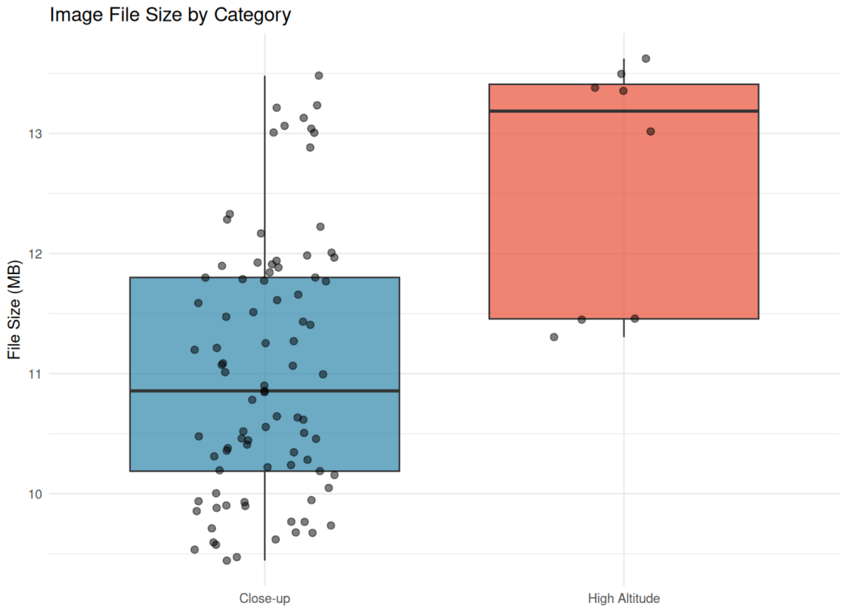

Figure 5: Distribution of image file sizes by category.

- Close-up images: mean 11.0 MB
- High-altitude images: mean 12.6 MB

4.6 Similarity Search Validation

Content-based image retrieval was tested using high-altitude images as queries. Results confirm that similarity search returns semantically consistent results:

Table 3: Top-5 similar images for query DJI_0552.JPG (high-altitude)

Rank	Image	Category
1	DJI_0552.JPG	High-altitude (query)
2	DJI_0557.JPG	High-altitude
3	DJI_0555.JPG	High-altitude
4	DJI_0556.JPG	High-altitude
5	DJI_0554.JPG	High-altitude

[FIND: All top-5 results for a high-altitude query are also high-altitude images, demonstrating effective semantic retrieval**]**

5 Discussion

5.1 Interpretation of Results

5.1.1 Semantic Clustering Without Domain-Specific Training

The most significant finding is that CLIP embeddings—trained on general web images, not agricultural data—successfully separated drone images by their functional purpose (inspection vs. mapping). This suggests that:

1. Pretrained vision-language models capture transferable visual concepts

2. Altitude-related visual features (scale, detail level, field-of-view) are well-represented in general visual embeddings

3. Domain-specific fine-tuning may not be necessary for basic organizational tasks

[OBS: CLIP's training on diverse web data appears to include sufficient exposure to aerial/satellite imagery to enable meaningful agricultural drone image understanding**]**

5.1.2 Embedding Space vs. Geographic Space

A key advantage of embedding-based organization over GPS metadata is the capture of *semantic* rather than *spatial* similarity. Images taken at different GPS locations but serving the same purpose (e.g., all high-altitude mapping shots) cluster together in embedding space, whereas GPS coordinates would scatter them across the field.

[TODO: Add quantitative comparison: GPS clustering vs. embedding clustering**]**

5.1.3 Cluster Cohesion Asymmetry

The high-altitude cluster showed higher cohesion (lower mean neighbor distance) than the close-up cluster. This may reflect:

- **Homogeneity of overview images**: High-altitude shots capture similar field patterns regardless of position
- **Diversity of inspection targets**: Close-up images may show varied crop conditions, growth stages, or inspection targets

[FIND: Cluster cohesion differences may indicate semantic diversity within categories—useful for quality assessment or anomaly detection**]**

5.2 Practical Implications

5.2.1 Automated Dataset Curation

The demonstrated clustering capability enables automated organization of drone survey data:

- Separate mapping images from inspection images automatically
- Identify outliers or unusual images for review
- Filter large datasets to specific image types without manual labeling

5.2.2 Content-Based Image Retrieval

Similarity search based on embeddings provides intuitive "find similar" functionality:

- Query by example: find all images similar to a selected sample
- Batch processing: process only images of a specific type
- Quality control: identify near-duplicates or redundant coverage

5.3 Limitations

1. **Sample size**: Analysis based on 97 images from a single field; generalization requires validation on larger, more diverse datasets
2. **Binary clustering**: Current analysis identified only two categories; finer-grained distinctions (e.g., crop health categories) were not explored
3. **UMAP projection**: 2D visualization may lose information present in the full 512D embedding space
4. **Model selection**: Only CLIP ViT-B/32 was evaluated; other models may perform differently

[TODO: Address limitations in future work section**]**

5.4 Comparison with Related Work

[TODO: Add comparison with existing drone image organization methods, GPS-based clustering, traditional image classification approaches**]**

5.5 Future Directions

1. **Multi-field validation**: Apply methodology to datasets from multiple agricultural sites
2. **Fine-grained classification**: Explore embeddings for crop health assessment, disease detection
3. **Temporal analysis**: Track embedding changes across growing season
4. **Model comparison**: Evaluate agricultural-specific models vs. general pretrained models

6 Conclusions

This study demonstrated that pretrained vision-language embeddings (CLIP) effectively organize agricultural drone imagery by semantic content without domain-specific training. Key findings include:

1. **Automatic clustering**: CLIP embeddings with UMAP projection separated 97 drone images into two semantically meaningful clusters: close-up crop inspection (89 images, 92%) and high-altitude field mapping (8 images, 8%).
2. **Strong cluster separation**: Only 2.7% of nearest-neighbor connections crossed category boundaries, indicating robust semantic clustering.
3. **Effective similarity search**: Content-based image retrieval correctly identified semantically similar images, enabling intuitive "find similar" functionality.
4. **Complementary to GPS**: Embedding-based organization captures semantic similarity (image purpose) orthogonal to spatial position captured by GPS metadata.

These results suggest that pretrained vision-language models offer a practical, zero-configuration approach for organizing large agricultural drone datasets. By reducing manual curation effort, this methodology can accelerate precision agriculture workflows and enable more efficient use of drone survey data.

[TODO: Refine conclusions after completing discussion section**]**

A Observation Log

Log Format

Each observation entry includes:

- **Date**: When the observation was made
- **Category**: Type of observation (data, method, result, idea)
- **Description**: The observation itself
- **Link**: Reference to relevant section, figure, or data

2026-01-13

OBS-001 [Data] Dataset loaded successfully: 97 DJI drone images from Santa Ines field, 10-12 MB each.

OBS-002 [Result] UMAP projection reveals clear bimodal structure—main cluster of 89 images and outlier cluster of 8 images at bottom-right of embedding space.

OBS-003 [Result] Outlier images (DJI_0469, DJI_0509, DJI_0552-0557) visually confirmed as high-altitude overview shots, distinct from close-up crop inspection images.

OBS-004 [Method] CLIP embeddings capture altitude/scale differences effectively—semantic clustering emerges without altitude metadata being provided.

2026-01-14

OBS-005 [Result] Similarity index computed successfully. Query tests confirm semantically consistent retrieval—high-altitude queries return only high-altitude neighbors.

OBS-006 [Result] Network analysis: 97.3% of k-NN edges stay within category. Only 8/291 edges cross category boundary.

OBS-007 [Result] Cluster cohesion asymmetry: high-altitude cluster is 35% tighter (mean neighbor distance 0.276 vs 0.424). May indicate greater visual homogeneity of overview shots.

OBS-008 [Idea] Cohesion metric could serve as diversity indicator—tight clusters = homogeneous content, diffuse clusters = varied content. Useful for coverage assessment.

OBS-009 [Method] R/ggplot2 workflow established for reproducible analysis. Scripts: `analysis.R`, `similarity_analysis.R`.

Template for Future Entries

OBS-XXX [Category] Description of observation. *See Section X / Figure X*

Index by Category

Data observations: OBS-001
Method observations: OBS-004, OBS-009
Result observations: OBS-002, OBS-003, OBS-005, OBS-006, OBS-007
Ideas/hypotheses: OBS-008

B Technical Details

B.1 Software Versions

Software	Version
FiftyOne	1.12.0
Python	3.12
R	4.3.3
PyTorch	(system version)
CLIP model	ViT-B/32

B.2 Data Paths

```
research/
+-- paper/
|   +-- main.tex
|   +-- sections/
|   |   +-- abstract.tex
|   |   +-- introduction.tex
|   |   +-- objectives.tex
|   |   +-- methods.tex
|   |   +-- results.tex
|   |   +-- discussion.tex
|   |   +-- conclusions.tex
|   |   +-- observations.tex
|   |   +-- technical.tex
|   +-- figures/
+-- data/
|   +-- raw/
|   +-- processed/
|       +-- santa_ines_data.csv
|       +-- similarity_neighbors.csv
+-- analysis/
|   +-- R/
|   |   +-- analysis.R
|   |   +-- similarity_analysis.R
|   +-- python/
+-- notes/
+-- logs/
```

B.3 Key Commands

B.3.1 FiftyOne Dataset Operations

```
import fiftyone as fo
import fiftyone.brain as fob

# Load dataset
dataset = fo.load_dataset("santa_ines_dron")

# Access embeddings
results = dataset.load_brain_results("clip_viz")
```

```
points = results.points  # Shape: (97, 2)

# Similarity search
view = dataset.sort_by_similarity(
    sample_id, k=10, brain_key="clip_similarity"
)

# Filter by tags
high_alt = dataset.match_tags("high-altitude")
closeups = dataset.match_tags("close-up")
```

B.3.2 Compilation

```
# Compile LaTeX document
cd research/paper
pdflatex main.tex
pdflatex main.tex  # Run twice for TOC

# Or use latexmk for automatic compilation
latexmk -pdf main.tex
```

B.4 FiftyOne Brain Keys

Brain Key	Type	Description
clip_viz	visualization	CLIP embeddings + UMAP 2D projection
clip_similarity	similarity	Similarity index for retrieval

B.5 GPS Extraction from Drone Images

GPS coordinates were extracted from EXIF metadata embedded in DJI drone images:

```
from PIL import Image
from PIL.ExifTags import TAGS, GPSTAGS

def get_gps_coords(image_path):
    """Extract GPS coordinates from EXIF data"""
    image = Image.open(image_path)
    exif_data = image._getexif()

    gps_info = exif_data.get('GPSInfo', {})

    # Convert DMS to decimal degrees
    def to_degrees(dms):
        d, m, s = dms
        return d + m/60.0 + s/3600.0

    lat = to_degrees(gps_info['GPSLatitude'])
    if gps_info['GPSLatitudeRef'] == 'S':
        lat = -lat

    lon = to_degrees(gps_info['GPSLongitude'])
    if gps_info['GPSLongitudeRef'] == 'W':
        lon = -lon
```

```
    return lat, lon
```

B.6 Satellite Map Integration

Google Maps Static API was used to obtain satellite imagery background:

```
import requests

API_KEY = "YOUR_API_KEY"
center_lat, center_lon = -34.4746, -70.9516
zoom = 18  # High zoom for agricultural detail

url = (f"https://maps.googleapis.com/maps/api/staticmap"
       f"?center={center_lat},{center_lon}"
       f"&zoom={zoom}&size=640x640"
       f"&maptype=satellite&key={API_KEY}")

response = requests.get(url)
satellite_img = plt.imread(BytesIO(response.content))
```

B.6.1 Coordinate Projection

GPS coordinates were converted to image pixel coordinates using Web Mercator projection:

```
import numpy as np

def gps_to_pixels(lat, lon, center_lat, center_lon,
                  zoom, img_size):
    """Convert GPS to image pixel coordinates"""
    def lat_lon_to_world(lat, lon, zoom):
        lat_rad = np.radians(lat)
        n = 2 ** zoom
        x = (lon + 180) / 360 * n * 256
        y = (1 - np.log(np.tan(lat_rad) +
             1/np.cos(lat_rad)) / np.pi) / 2 * n * 256
        return x, y

    cx, cy = lat_lon_to_world(center_lat, center_lon, zoom)
    px, py = lat_lon_to_world(lat, lon, zoom)

    x = (px - cx) + img_size / 2
    y = (py - cy) + img_size / 2
    return x, y
```

B.7 Analysis Scripts Reference

Table 4: Python scripts for analysis pipeline

Script	Purpose
gps_satellite_map.py	Extract GPS from EXIF metadata
satellite_map.py	Generate satellite map with GPS overlay
cluster_samples.py	Select representative images per cluster
combined_figure.py	Create 4-panel combined analysis figure

Table 5: R scripts for statistical analysis

Script	**Purpose**
analysis.R	Embedding visualization, cluster statistics
similarity_analysis.R	Network analysis, cohesion metrics